SUPPLEMENTARY WORKSHEET: WEIERSTRASS CUBICS

This worksheet supplements the previous lecture on elliptic curves. We have already discussed the construction of a Weierstrass equation from the Riemann–Roch theorem, the chord–tangent group law, the explicit addition and doubling formulas, and the fact that the group operations are morphisms. We will not repeat those facts here. Instead, we collect the algebraic quantities attached to a Weierstrass equation, their behaviour under changes of variables, and a few useful facts about singular Weierstrass cubics.

1. The standard quantities attached to a Weierstrass equation

Let

$$E: \quad y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6 \tag{1.1}$$

be a Weierstrass equation over a field K. The coefficients $a_1,\ a_2,\ a_3,\ a_4,\ a_6$ are sometimes called the *a-invariants* of the equation.

Definition 1.1. Associated to (1.1), define

$$\begin{aligned} b_2 &= a_1^2 + 4a_2, \\ b_4 &= a_1a_3 + 2a_4, \\ b_6 &= a_3^2 + 4a_6, \\ b_8 &= a_1^2a_6 + 4a_2a_6 - a_1a_3a_4 + a_2a_3^2 - a_4^2. \end{aligned}$$

Then define

$$\begin{aligned} c_4 &= b_2^2 - 24b_4, \\ c_6 &= -b_2^3 + 36b_2b_4 - 216b_6, \end{aligned}$$

and

$$\Delta = -b_2^2b_8 - 8b_4^3 - 27b_6^2 + 9b_2b_4b_6.$$

The quantity Δ is called the *discriminant* of the Weierstrass equation. If $\Delta \neq 0$, define its *j-invariant* by

$$j(E) = \frac{c_4^3}{\Delta}.$$

The following identities are useful both for checking computations and for remembering the definitions:

$$b_2b_6 - b_4^2 = 4b_8, \tag{1.2}$$

$$c_4^3 - c_6^2 = 1728\Delta. \tag{1.3}$$

The first identity is valid without dividing by 4, so it remains useful in characteristic 2. Similarly, the definition of Δ above makes sense in every characteristic.

Remark 1.2 (Completing the square). A direct calculation gives

$$(2y + a_1x + a_3)^2 = 4x^3 + b_2x^2 + 2b_4x + b_6. \tag{1.4}$$

Thus the b-invariants package the cubic polynomial that appears after completing the square. This formula is valid in every characteristic, even though it can only be used as a genuine change of variables when 2 is invertible in K.

Recall that $-(x, y) = (x, -y - a_1x - a_3)$. Consequently, an affine point $P = (x, y)$ satisfies $[2]P = O$ if and only if

$$2y + a_1x + a_3 = 0.$$

Combining this with (1.4), the x-coordinates of the nonzero points of order 2 are precisely the roots of

$$4x^3 + b_2x^2 + 2b_4x + b_6.$$

2. Admissible changes of variables

We have already seen that two Weierstrass equations for the same pointed elliptic curve are related by an admissible change of variables

$$x = u^2x' + r, \qquad y = u^3y' + su^2x' + t, \tag{2.1}$$

where $u \in K^*$ and $r, s, t \in K$. Let a_i' denote the coefficients of the equation in the variables x', y'.

Proposition 2.1 (Change-of-variable formulas)**.** *Under* (2.1)*, the coefficients transform according to*

$$\begin{aligned}
ua_1' &= a_1 + 2s,\\
u^2a_2' &= a_2 - sa_1 + 3r - s^2,\\
u^3a_3' &= a_3 + ra_1 + 2t,\\
u^4a_4' &= a_4 - sa_3 + 2ra_2 - (t + rs)a_1 + 3r^2 - 2st,\\
u^6a_6' &= a_6 + ra_4 + r^2a_2 + r^3 - ta_3 - t^2 - rta_1.
\end{aligned}$$

The derived quantities satisfy

$$\begin{aligned}
u^2b_2' &= b_2 + 12r,\\
u^4b_4' &= b_4 + rb_2 + 6r^2,\\
u^6b_6' &= b_6 + 2rb_4 + r^2b_2 + 4r^3,\\
u^8b_8' &= b_8 + 3rb_6 + 3r^2b_4 + r^3b_2 + 3r^4,
\end{aligned}$$

and, most importantly,

$$u^4c_4' = c_4, \qquad u^6c_6' = c_6, \qquad u^{12}\Delta' = \Delta, \qquad j' = j. \tag{2.2}$$

For the differential introduced earlier,

$$\omega = \frac{dx}{2y + a_1x + a_3}, \qquad \omega' = \frac{dx'}{2y' + a_1'x' + a_3'},$$

one has

$$\omega = u^{-1}\omega'.$$

Sketch of proof. Substitute (2.1) into the original equation and compare coefficients of the monomials in x' and y'. This gives the five formulas for the a_i'. Substitution into the definitions of b_i, c_i and Δ gives the remaining formulas. Similar arguement works for differentials. □

3. The discriminant and singular cubics

Proposition 3.1. *Let C be the projective cubic defined by a Weierstrass equation.*

(i) The curve C is nonsingular if and only if $\Delta \neq 0$.
(ii) If $\Delta = 0$ and $c_4 \neq 0$, then C has a node.
(iii) If $\Delta = c_4 = 0$, then C has a cusp.

Sketch of proof. A singular point must satisfy the projective equation and all three of its first partial derivatives. Eliminating the coordinates from these equations gives precisely the condition $\Delta = 0$. When a singular point exists, the quadratic part of the local equation is the tangent cone. It factors as two distinct lines exactly when $c_4 \neq 0$, giving a node, and as a repeated line when $c_4 = 0$, giving a cusp. In characteristic different from 2, the first assertion can also be seen from (1.4): singularity is equivalent to the cubic on the right having a repeated root. □

Thus the same polynomial expression Δ has two related uses:

- $\Delta \neq 0$ is the condition that the Weierstrass cubic is an elliptic curve;
- the scaling rule $u^{12}\Delta' = \Delta$ records how the discriminant depends on the chosen Weierstrass coordinates.

In particular, the value of Δ is attached to an equation, whereas its vanishing or nonvanishing is attached to the curve.

4. Short Weierstrass equations

Assume first that $\mathrm{char}(K) \neq 2$. Replacing $2y + a_1x + a_3$ by a new y-coordinate puts the equation into the form

$$y^2 = 4x^3 + b_2x^2 + 2b_4x + b_6.$$

If in addition $\mathrm{char}(K) \neq 3$, translating the x-coordinate removes the quadratic term. After dividing the y-coordinate by 2, one obtains a short Weierstrass equation

$$E: \quad y^2 = x^3 + Ax + B. \tag{4.1}$$

One may take

$$A = -\frac{c_4}{48}, \qquad B = -\frac{c_6}{864}.$$

For (4.1), all the standard quantities become

$$b_2 = 0, \qquad b_4 = 2A, \qquad b_6 = 4B, \qquad b_8 = -A^2,$$
$$c_4 = -48A, \qquad c_6 = -864B, \qquad \Delta = -16(4A^3 + 27B^2).$$

Hence

$$j(E) = 1728\frac{4A^3}{4A^3 + 27B^2}. \tag{4.2}$$

The curve is nonsingular exactly when

$$4A^3 + 27B^2 \neq 0.$$

5. The j-invariant

The transformation formula (2.2) shows that $j(E)$ is unchanged by every admissible change of Weierstrass coordinates. It is therefore an invariant of the elliptic curve, rather than of a particular equation.

Theorem 5.1. *Let E_1 and E_2 be elliptic curves over K.*

(i) If E_1 and E_2 are isomorphic over K, then

$$j(E_1) = j(E_2).$$

(ii) The curves E_1 and E_2 are isomorphic over $\bar{K}$ if and only if

$$j(E_1) = j(E_2).$$

Sketch of proof. The first assertion follows immediately from the change-of-variable formulas. Conversely, after extending scalars to $\bar{K}$, one uses admissible changes of variables to put both curves into standard forms determined by their common value of j. The exceptional values require separate normal forms, but the conclusion remains the same in every characteristic. □

Equality of j-invariants need not imply isomorphism over the original field. Curves with the same j-invariant which become isomorphic only after extending the field are called *twists* of one another. For example, when $\mathrm{char}(K) \neq 2, 3$, the curves

$$y^2 = x^3 - x \qquad \text{and} \qquad y^2 = x^3 - d^2x$$

have $j = 1728$ and become isomorphic over $\bar{K}$, but need not be isomorphic over K.

For a nonsingular short Weierstrass equation in characteristic different from 2 and 3, formula (4.2) gives

$$j(E) = 0 \quad \Longleftrightarrow \quad A = 0,$$

and

$$j(E) = 1728 \quad \Longleftrightarrow \quad B = 0.$$

6. A supplement from the group law: singular cubics

The chord–tangent construction is not restricted to nonsingular cubics. If a Weierstrass cubic is singular, its nonsingular points still form an abelian group with identity O. These groups are much simpler than elliptic curves.

Proposition 6.1. *Let C be an irreducible singular Weierstrass cubic over $\bar{K}$, and let C_{ns} denote its nonsingular points.*

(i) If C has a node, then

$$C_{\mathrm{ns}} \cong \bar{K}^*$$

as abelian groups.

(ii) If C has a cusp, then

$$C_{\mathrm{ns}} \cong (\bar{K}, +)$$

as abelian groups.

Here are two concrete models, assuming $\mathrm{char}(K) \neq 2$.

Example 6.2 (A nodal cubic)**.** Consider $C : y^2 = x^2(x+1)$.

Its singular point is $(0,0)$. A line through the singular point with slope t meets the curve again at

$$(x, y) = (t^2 - 1, t(t^2 - 1)).$$

On the nonsingular locus, the function

$$u = \frac{y - x}{y + x}$$

takes values in $\bar{K}^*$, sends O to 1, and identifies the chord–tangent law with multiplication. Indeed, negation sends u to u^{-1}, and three collinear nonsingular points have u-parameters whose product is 1.

Example 6.3 (A cuspidal cubic)**.** Consider $C : y^2 = x^3$.

Its singular point is again $(0,0)$. On the nonsingular locus, the function

$$u = \frac{x}{y}$$

extends by $u(O) = 0$ and identifies the chord–tangent law with addition in $\bar{K}$. Three collinear nonsingular points have u-parameters whose sum is 0.

This explains geometrically why a singular cubic is not an elliptic curve: after normalization, its group of nonsingular points is an additive or multiplicative one-dimensional algebraic group rather than a projective genus-one group.

7. Quick checks

Exercise 7.1. For each of the following equations, compute c_4, c_6, Δ and j, and determine whether the cubic is nonsingular:

$$E_1 : y^2 = x^3 - x, \qquad E_2 : y^2 = x^3 + 1, \qquad E_3 : y^2 = x^3 - 3x + 2.$$

If a curve is singular, decide whether it is nodal or cuspidal.

Exercise 7.2. Verify directly the identities (1.2) and (1.3) from the definitions. The second computation is longer, but it is a useful consistency check for the constants $24, 36, 216$ and 1728.

Exercise 7.3. Assume $\mathrm{char}(K) \neq 2, 3$. Let

$$E : y^2 = x^3 + Ax + B$$

and make the change of variables

$$x = u^2 x', \qquad y = u^3 y'.$$

Compute A', B', Δ' and verify directly that $j(E') = j(E)$.